Aaryan S. Maralihalli

Machine Learning Engineer — Edge & Multimodal AI

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Summary

ML engineer who builds **small, efficient deep models on raw sensor streams** — an IEEE-published **~238K-parameter 1D-CNN** over multi-channel ECG/PPG waveforms (ROC-AUC 0.757), multimodal fusion, and **fully-local LLM systems with sub-100ms retrieval** (no cloud dependency). Comfortable from self-supervised training through **FastAPI/Docker deployment** and real-time pipelines — the model-to-edge path. CS grad, Jul 2026.

Experience

HyperVerge, Bangalore
Integration Engineer Intern Jan 2026 – Jun 2026
AI identity-verification platform — production computer-vision modules

- Integrated **production vision/AI verification modules** (face match, liveness/selfie, document OCR) for **6 enterprise fintech clients**; took **3 live at >95% verification conversion — a 10% lift** — with post-go-live performance monitoring and edge-case failure analysis.
- Authored Solution Design Docs and drove model-flow tuning against real-world capture conditions (drop-off and conversion diagnostics across 15+ lines of business).

Integrated Genetic Solutions, Bangalore
Project Intern, Medical AI Jun 2024 – Aug 2024

- Worked on ML workflows for medical document extraction on a HIPAA-compliant product; led a legacy-JS → **React (MERN)** migration improving processing of **5M+ lab values**.

Selected Projects

Self-Supervised Sensor-Stream Model (IEEE-published) github.com/Aaryansm11/ecg-ppg-arrhythmia-prediction

- **Contrastive learning (SimCLR/NT-Xent)** on **MIMIC-III** raw ECG/PPG — **133K segments, 8 channels**; designed a custom **1D-CNN WaveformEncoder (~238K params, 128-d embeddings)** — compact enough for constrained inference; **multimodal fusion** with a clinical Random Forest; **ROC-AUC 0.757** vs CHA2DS2-VASc baseline; **XAI** (temporal saliency, channel importance, integrated gradients); FastAPI inference server.

LANGTRON — fully-local LLM system github.com/Aaryansm11/LANGTRON

- Fine-tuned **Mistral-7B (LoRA)**; dual-model architecture; **FAISS retrieval at sub-100ms**; runs **100% on-device, zero cloud** — built for the privacy/latency constraints edge AI lives under.

FinFusion — multimodal forecasting

- **LSTM time-series + transformer sentiment fusion**, improving MAPE 15–20% over single-modal baselines.

Live Emostream — real-time stream processing github.com/Aaryansm11/Live_emostream

- Event-driven **Kafka + Spark + Redis** pipeline with nginx load balancing; **load-tested to 1,000 concurrent users** (Locust).

Langtron-KGS github.com/Aaryansm11/Langtron-KGS

- SpaCy NER + BERT relation extraction → **Neo4j** knowledge graph; FastAPI microservice.

Skills

- **ML / DL:** PyTorch, contrastive/self-supervised learning, CNN architecture design (1D/temporal), multimodal fusion, LoRA fine-tuning, HuggingFace, XAI (saliency, integrated gradients), scikit-learn
- **Efficiency & Serving:** compact-model design, local/offline LLM inference (FAISS, quantized runtimes), FastAPI model serving, Docker, AWS, streaming inference (Kafka, Redis, Spark)
- **Foundations:** Python, SQL, Java, C-family; signal/time-series processing (HDF5, NumPy/Pandas); **Graphics Programming with WebGL** (PESU I/O certification); DSA (150+ LeetCode)

Publication & Education

- **IEEE ICASI 2026 (Kyoto), Oral Presentation** — *"An Interpretable Model for Stroke Prediction and Arrhythmia Classification using ECG & PPG."*
- **B.Tech, Computer Science** — PES University, Bangalore — 2022 – Jul 2026